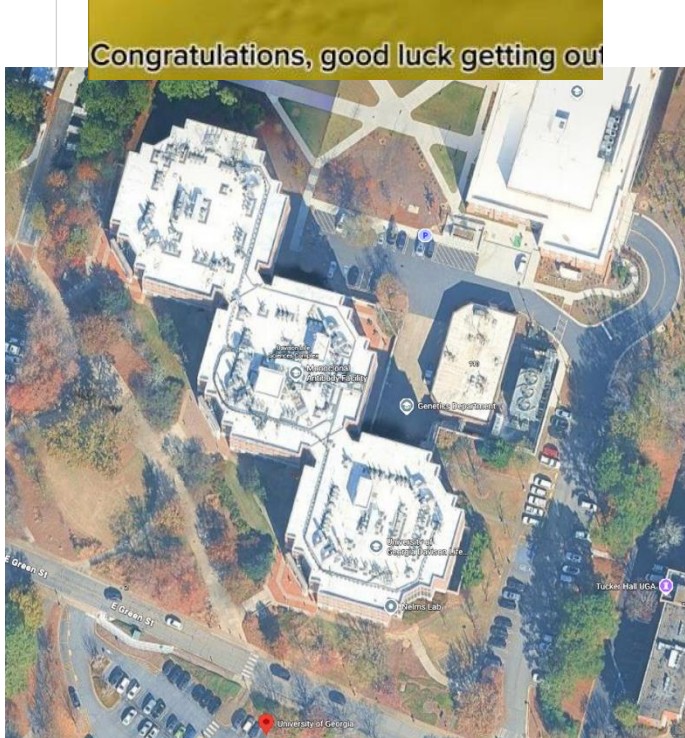


ISE 430/530: Manufacturing Systems and Facilities Design



Dr. Cheng (Erick) Chen
Industrial & Systems and Engineering Management
Department

Cheng.chen@uah.edu

Monoclonal Antibody Facility

Group Technology

- Group Technology (Product Family, Cellular manufacturing)
 - "A manufacturing philosophy that identifies and exploits the similarity of parts and processes in design and manufacture."
 - Aggregating medium volume-variety parts into families based on manufacturing operations or design attributes, and processing them within machine group ("cell")
 - Applicable in batch manufacturing - medium sized lots.
 - Advantages:
 - Shorter production time
 - Better production planning and control - small cells
 - Reduced tooling and set-up
 - Easy retrieval of design & process planning - variant mode

Benefits of GT and CMS (Companies Reporting)

- 70% Report reduced throughput time (even more report better predictability of delivery)
- 82% Report reduced numbers of overdue orders
- 42% Report reduced raw-materials inventory
- 62% Report reduced WIP
- 60% Report reduced finished goods inventory
- 33% Report increased employee output/time unit (productivity improvement)

Cell Formation Methods

- Direct clustering algorithm (DCA)

Part#\ Machine#	A	B	C	D	E	# of 1s
1	1		1			2
2	1					1
3		1		1	1	3
4	1		1			2
5		1				1
6				1	1	2
# of 1s	3	2	2	2	2	

Machine-Part Matrix

Directed Clustering Algorithm

- **Step 1:** Order the rows and columns using the sum of 1s in each column and row of machine-part matrix. Order rows (top to bottom) in descending; columns (left – right) in ascending. Where ties exist, break the ties in ascending/descending (be consistent if an option is picked) numerical sequence.
- **Step 2:** Sort the columns. Beginning with the first row of the matrix, shift to the left of the matrix all columns having a 1 in the first row. Continue the process row by row until no further opportunity exists for shifting columns.
- **Step 3:** Sort the rows. Column by column, beginning with the leftmost column, shift rows upward when opportunities exist to form blocks of 1s.
- **Step 4:** Form cells

Example-Cell Formation

- Example 1: DCA method to group 5 machines

Part#\ Machine#	A	B	C	D	E	# of 1s
1	1		1			2
2	1					1
3		1		1	1	3
4	1		1			2
5		1				1
6				1	1	2
# of 1s	3	2	2	2	2	

STEP 1-1: Sorting by rows in decreasing order

Part#\ Machine#	A	B	C	D	E	# of 1s
3		1		1	1	3
1	1		1			2
4	1		1			2
6				1	1	2
2	1					1
5		1				1
# of 1s	3	2	2	2	2	

Group Technology

- Example 1: DCA method to group 5 machines

Part#\ Machine#	A	B	C	D	E	# of 1s
3		1		1	1	3
1	1		1			2
4	1		1			2
6				1	1	2
2	1					1
5		1				1
# of 1s	3	2	2	2	2	

STEP 1-2: Sorting by columns in increasing order

Part#\ Machine#	E	D	C	B	A	# of 1s
3	1	1		1		3
1			1		1	2
4			1		1	2
6	1	1				2
2					1	1
5				1		1
# of 1s	2	2	2	2	3	

Example-Cell Formation

- Example 1: DCA method to group 5 machines

Part#\ Machine#	E	D	C	B	A	# of 1s
3	1	1		1		3
1			1		1	2
4			1		1	2
6	1	1				2
2					1	1
5				1		1
# of 1s	2	2	2	2	3	

Step 2: Beginning with the first row of the matrix, shift to the left of the matrix all columns having a 1 in the first row. Continue the process row by row until no further opportunity exists for shifting columns.

STEP 2: Sorting columns

(all columns have 1 in the first row. Continue row by row)

Part#\ Machine#	E	D	B	C	A	# of 1s
3	1	1	1			3
1				1	1	2
4				1	1	2
6	1	1				2
2					1	1
5			1			1
# of 1s	2	2	2	2	3	

Example-Cell Formation

- Example 1: DCA method to group 5 machines

Part#\ Machine#	E	D	B	C	A	# of 1s
3	1	1	1			3
1				1	1	2
4				1	1	2
6	1	1				2
2					1	1
5			1			1
# of 1s	2	2	2	2	3	

Step 3: Sort the rows. Column by column, beginning with the leftmost column, shift rows upward when opportunities exist to form blocks of 1s.

STEP 3: Sorting rows

(all rows have 1 in the first column. Continue column by column)

Part#\ Machine#	E	D	B	C	A	# of 1s
3	1	1	1			3
6	1	1				2
5			1			1
1				1	1	2
4				1	1	2
2					1	1
# of 1s	2	2	2	2	3	

Example-Cell Formation

- Example 1: DCA method to group 5 machines

Part#\ Machine#	E	D	B	C	A	# of 1s
3	1	1	1			3
6	1	1				2
5			1			1
1				1	1	2
4				1	1	2
2					1	1
# of 1s	2	2	2	2	3	

STEP 4: Form cells

Group machines B, D, E and process parts 3, 5, 6 in this area,

Group machines A, C and process parts 1, 2, 4 in this area.

Example-Cell Formation

- Example 2: Issues with forming cells

Part#\ Machine#	E	D	B	C	A
3	1	1	1		
6	1	1			
5			1		
1				1	1
4				1	1
2					1

Part#\ Machine#	E	D	B	C	A
3	1	1			
6	1	1			
5			1	1	
1				1	1
4				1	1
2					1

Cell formation is not perfect

Example-Cell Formation

- Example 2: Issues with forming cells

Part#\ Machine#	E	D	B	C	A
3	1	1	1		
6	1	1			
5			1	1	
1				1	1
4				1	1
2					1

For complex problems, the solution for DCA is not unique!

Part#\ Machine#	E	D	B ₁	B ₂	C	A
3	1	1	1			
6	1	1				
5				1	1	
1					1	1
4					1	1
2						1

Part#\ Machine#	E	D	B	C ₁	C ₂	A
3	1	1	1			
6	1	1				
5			1	1		
1					1	1
4					1	1
2						1

Example-Cell Formation

- Python Programming: (What if you swap step 2 with step 3?)
- <https://colab.research.google.com/drive/18lrk8neYUdLGOgOKcKw4LzUJNd0Wiwuz?usp=sharing>

Example-Cell Formation

- Example 3: Clustering 26 machines

Part #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	# of 1s
1	1		1		1			1	1		1						1	1				1					9
2	1		1		1			1	1		1						1	1				1					9
3	1		1		1			1	1		1						1	1				1					9
4	1		1		1			1	1		1						1	1				1					9
5	1		1			1		1	1		1						1	1				1					9
6	1		1			1		1	1		1						1	1				1					9
7	1		1			1		1	1		1						1	1				1					9
8		1		1			1			1		1							1	1				1			8
9		1		1			1			1		1							1	1				1			8
10	1		1										1	1	1	1					1			1	1	1	10
11	1		1										1	1	1	1					1			1	1	1	10
12	1		1										1	1	1	1					1			1	1	1	10
13	1		1										1	1	1	1					1			1	1	1	10
# of 1s	11	2	11	2	4	3	2	7	7	2	7	2	4	4	4	4	7	7	2	2	4	7	2	4	4	4	

Example-Cell Formation

- Example 3: Clustering 26 machines
- **Step 1:** Sorting by columns/rows in increasing/decreasing order

Part #	Machine #																										# of 1s
	23	20	19	12	10	7	4	2	6	26	25	24	21	16	15	14	13	5	22	18	21	17	11	9	3	1	
13										1	1	1	1	1	1	1	1								1	1	10
12										1	1	1	1	1	1	1	1								1	1	10
11										1	1	1	1	1	1	1	1								1	1	10
10										1	1	1	1	1	1	1	1								1	1	10
7									1										1	1	1	1	1	1	1	1	9
6									1										1	1	1	1	1	1	1	1	9
5									1										1	1	1	1	1	1	1	1	9
4																		1	1	1	1	1	1	1	1	1	9
3																		1	1	1	1	1	1	1	1	1	9
2																		1	1	1	1	1	1	1	1	1	9
1																		1	1	1	1	1	1	1	1	1	9
9	1	1	1	1	1	1	1	1																			8
8	1	1	1	1	1	1	1	1																			8
# of 1s	2	2	2	2	2	2	2	2	3	4	4	4	4	4	4	4	4	4	7	7	7	7	7	7	11	11	

Example-Cell Formation

- Example 3: Clustering 26 machines
- **Step 2:** Sorting by columns to fill the most upper rows

Example-Cell Formation

- Example 3: Clustering 26 machines
- **Step 3:** Sorting by rows to fill the most left columns

	Machine #																										
Part #	3	1	26	25	24	21	16	15	14	13	22	18	17	11	9	8	6	5	23	20	19	12	10	7	4	2	# of 1s
13	1	1	1	1	1	1	1	1	1	1																	10
12	1	1	1	1	1	1	1	1	1	1																	10
11	1	1	1	1	1	1	1	1	1	1																	10
10	1	1	1	1	1	1	1	1	1	1																	10
7	1	1									1	1	1	1	1	1	1										9
6	1	1									1	1	1	1	1	1	1										9
5	1	1									1	1	1	1	1	1	1										9
4	1	1									1	1	1	1	1	1		1									9
3	1	1									1	1	1	1	1	1		1									9
2	1	1									1	1	1	1	1	1		1									9
1	1	1									1	1	1	1	1	1		1									9
9																			1	1	1	1	1	1	1	1	8
8																			1	1	1	1	1	1	1	1	8
# of 1s	11	11	4	4	4	4	4	4	4	4	7	7	7	7	7	7	3	4	2	2	2	2	2	2	2	2	

Example-Cell Formation

- Example 3: Clustering 26 machines
- Step 4: From cells

Part #	Machine #																										# of 1s
	3	1	26	25	24	21	16	15	14	13	22	18	17	11	9	8	6	5	23	20	19	12	10	7	4	2	
13	1	1	1	1	1	1	1	1	1	1																	10
12	1	1	1	1	1	1	1	1	1	1																	10
11	1	1	1	1	1	1	1	1	1	1																	10
10	1	1	1	1	1	1	1	1	1	1																	10
7	1	1									1	1	1	1	1	1	1										9
6	1	1									1	1	1	1	1	1	1										9
5	1	1									1	1	1	1	1	1	1										9
4	1	1									1	1	1	1	1	1		1									9
3	1	1									1	1	1	1	1	1		1									9
2	1	1									1	1	1	1	1	1		1									9
1	1	1									1	1	1	1	1	1		1									9
9																			1	1	1	1	1	1	1	1	8
8																			1	1	1	1	1	1	1	1	8
# of 1s	11	11	4	4	4	4	4	4	4	4	7	7	7	7	7	7	3	4	2	2	2	2	2	2	2	2	

Example-Cell Formation

- Example 4: Clustering 26 machines

Part #	Machine #																										# of 1s
	3	1	26	25	24	21	16	15	14	13	22	18	17	11	9	8	6	5	23	20	19	12	10	7	4	2	
13	1	1	1	1	1	1	1	1	1	1																	10
12	1	1	1	1	1	1	1	1	1	1																	10
11	1	1	1	1	1	1	1	1	1	1																	10
10	1	1	1	1	1	1	1	1	1	1																	10
7	1	1									1	1	1	1	1	1	1										9
6	1	1									1	1	1	1	1	1	1										9
5	1	1									1	1	1	1	1	1	1										9
4	1	1									1	1	1	1	1	1			1								9
3	1	1									1	1	1	1	1	1			1								9
2	1	1									1	1	1	1	1	1			1								9
1	1	1									1	1	1	1	1	1			1								9
9																			1	1	1	1	1	1	1	1	8
8																			1	1	1	1	1	1	1	1	8
# of 1s	11	11	4	4	4	4	4	4	4	4	7	7	7	7	7	7	3	4	2	2	2	2	2	2	2	2	

||
Machines 1, 3:
"bottleneck"

Example-Cell Formation

- Example 4: Clustering 26 machines

Split bottleneck machines

split bottleneck machines

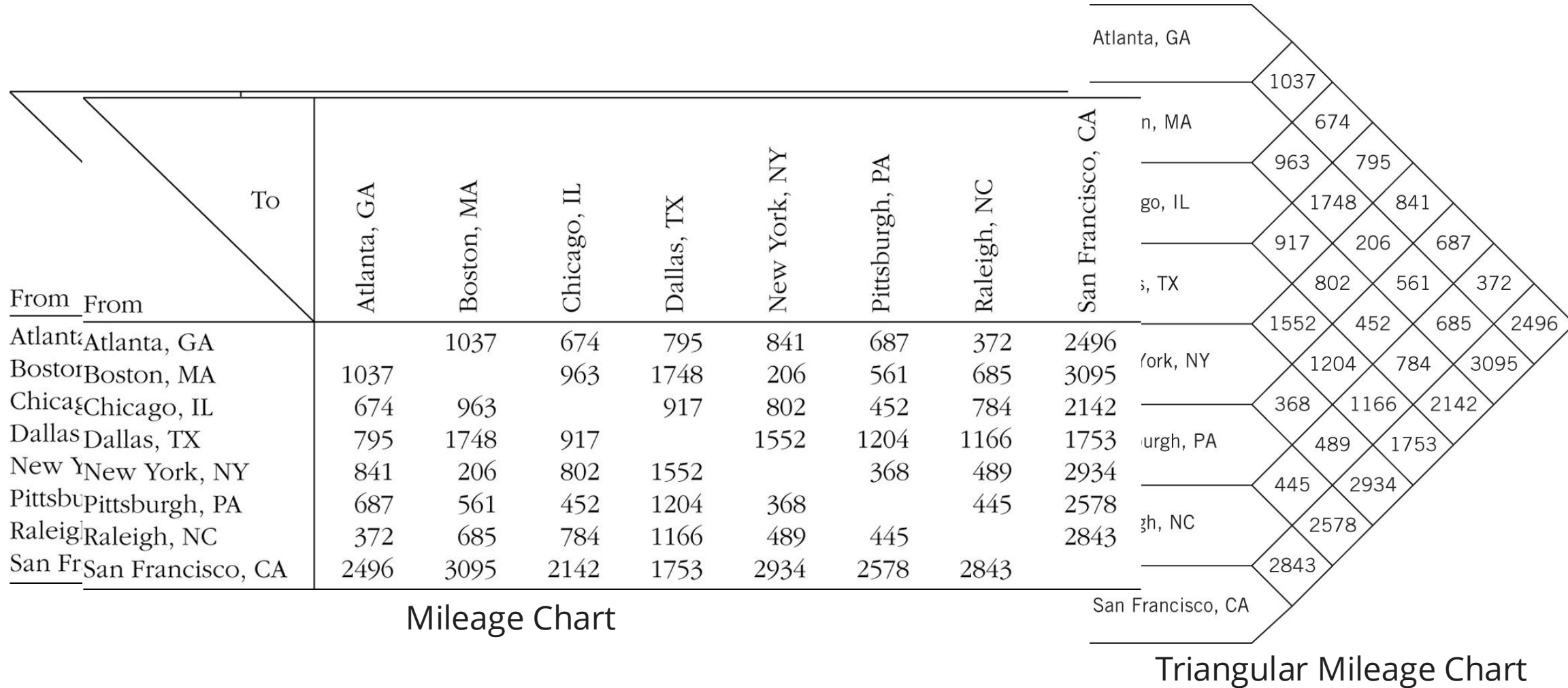
Part #	26	25	24	21	16	15	14	13	3a	1a	22	18	17	11	9	8	3b	1b	6	5	23	20	19	12	10	7	4	2						
13	1	1	1	1	1	1	1	1	1	1																								
12	1	1	1	1	1	1	1	1	1	1																								
11	1	1	1	1	1	1	1	1	1	1																								
10	1	1	1	1	1	1	1	1	1	1																								
7	Cell A										1	1	1	1	1	1	1	1	1															
6											1	1	1	1	1	1	1	1	1	1														
5											1	1	1	1	1	1	1	1	1	1														
4											1	1	1	1	1	1	1	1	1							1								
3											1	1	1	1	1	1	1	1	1							1								
2											1	1	1	1	1	1	1	1	1							1								
1											1	1	1	1	1	1	1	1	1							1								
9										Cell B																	Cell C							
8																																		

(c)

Activity Relationships

- Measuring the activities among departments is one of the most important elements in the layout of departments within a facility.
- In facilities having large volumes of materials, information, and people moving between departments, a quantitative measure of flow is important.
- In facilities having very little actual movement of materials, information, and people flowing between departments, but having significant communication and organizational interrelations, a qualitative measure of flow is important.

Quantitative Flow Measurement



Quantitative Flow Measurement

- From-to chart

To \ From	To						
	Stores	Milling	Turning	Press	Plate	Assembly	Warehouse
Stores		12	6	9	1	4	
Milling					7	2	
Turning		3			4		
Press					3	1	1
Plate		3	1			4	3
Assembly	1						7
Warehouse							

Note: need to use the same unit for each entry in the From-to chart.

Quantitative Flow Measurement

- Example 1. Developing From-to chart
 - A firm procures 3 components. Components 1 and 2 have same size and weight. Component 3 is twice as large. (i.e., moving 1 unit of component 3 is equivalent to moving 2 unit of component 1 or 2).
 - Departments A, B, C, D, E are in the facility. They are arranged in a linear flow.

Component	Production Quantities (per day)	Movement Factor	Equivalent Flows (per day)	Routing
1	30	1	30	A-C-B-D-E
2	12	1	12	A-B-D-E
3	7	2	14	A-C-D-B-E

Quantitative Flow Measurement

Component	Production Quantities (per day)	Movement Factor	Equivalent Flows (per day)	Routing
1	30	1	30	A-C-B-D-E
2	12	1	12	A-B-D-E
3	7	2	14	A-C-D-B-E

Construction:

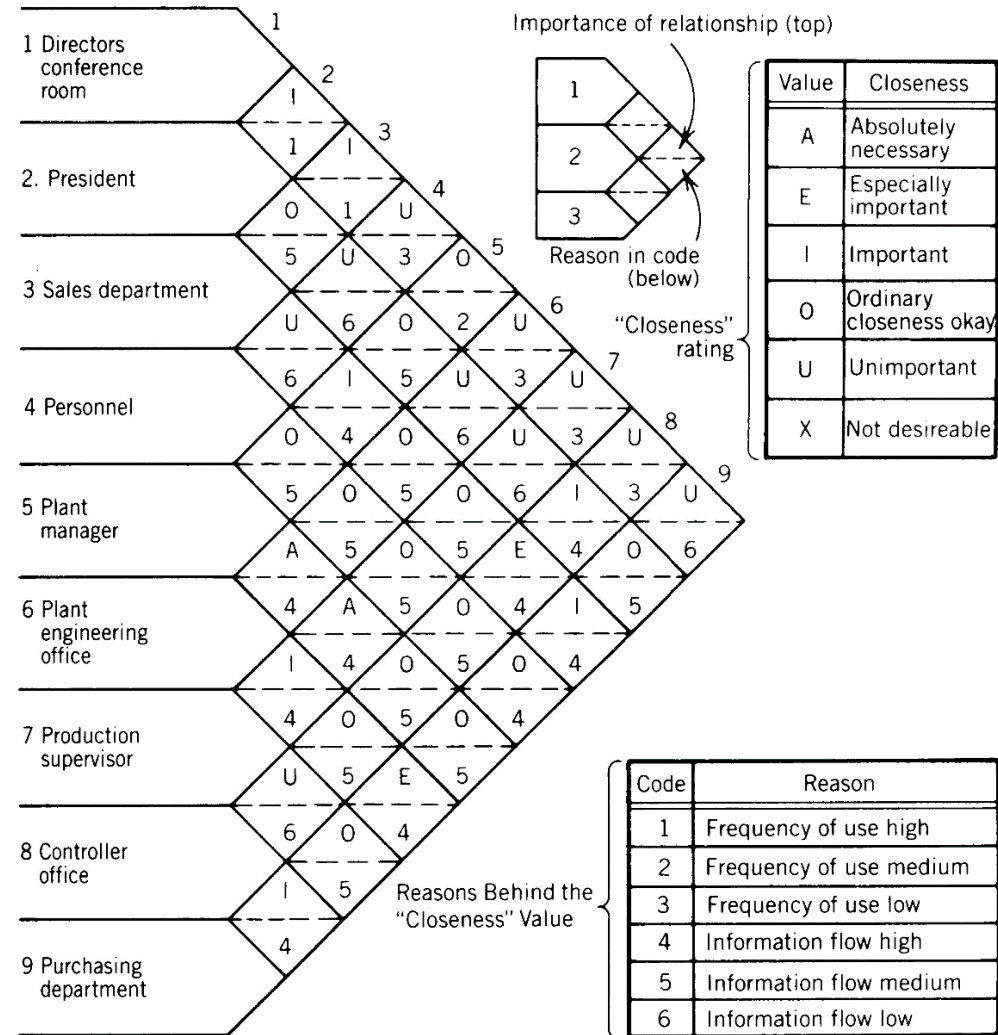
1. List all departments down the row and across the column.
2. Establish measure of flow for the facility that indicate equivalent flow volumes with respect to movement factor
3. Based on flow paths for items and established measure of flow, write flow volumes in from-to chart

To From	A	C	B	D	E
A		44	12	0	0
C	0		30	14	0
B	0	0		42	14
D	0	0	14		42
E	0	0	0	0	

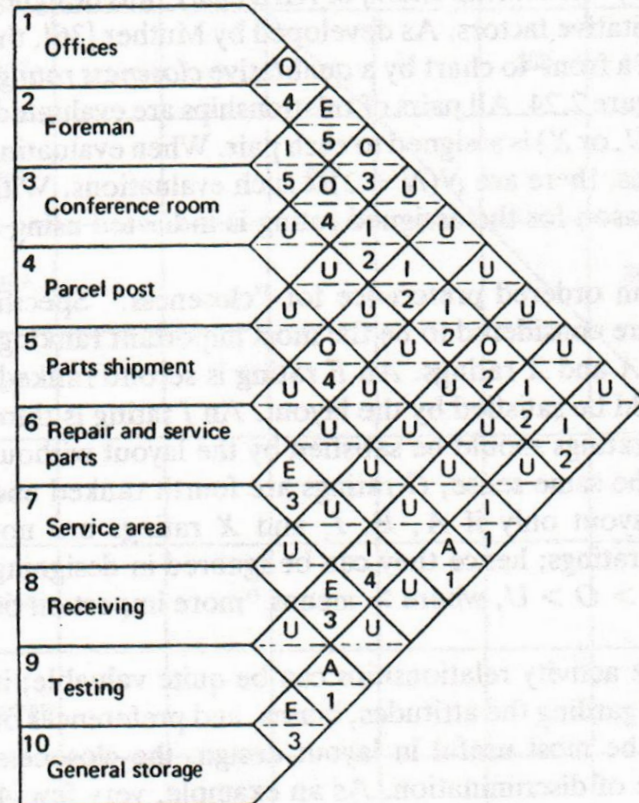
Activity Relationship Analysis

- Activity Relationship Chart
 - Shows the closeness ratings and reasons for closeness requirements between activities.
 - Construction procedure
 - 1) List all departments on the relationship chart
 - 2) Conduct interviews and surveys
 - 3) Define criteria for assigning closeness relationship (reasons for relationships)
 - 4) Establish relationship value and reason for the value
 - A: absolutely necessary,
 - E: especially important,
 - I: important,
 - O: ordinary closeness,
 - U: unimportant,
 - X: undesirable
 - 5) General review and discussion

Relationship Chart



Relationship Chart



Code	Reason
1	Flow of materials
2	Ease of supervision
3	Common personnel
4	Contact necessary
5	Convenience
6	
7	
8	
9	
10	

Rating	Definition
A	Absolutely necessary
E	Especially Important
I	Important
O	Ordinary closeness OK
U	Unimportant
X	Undesirable

Group Technology

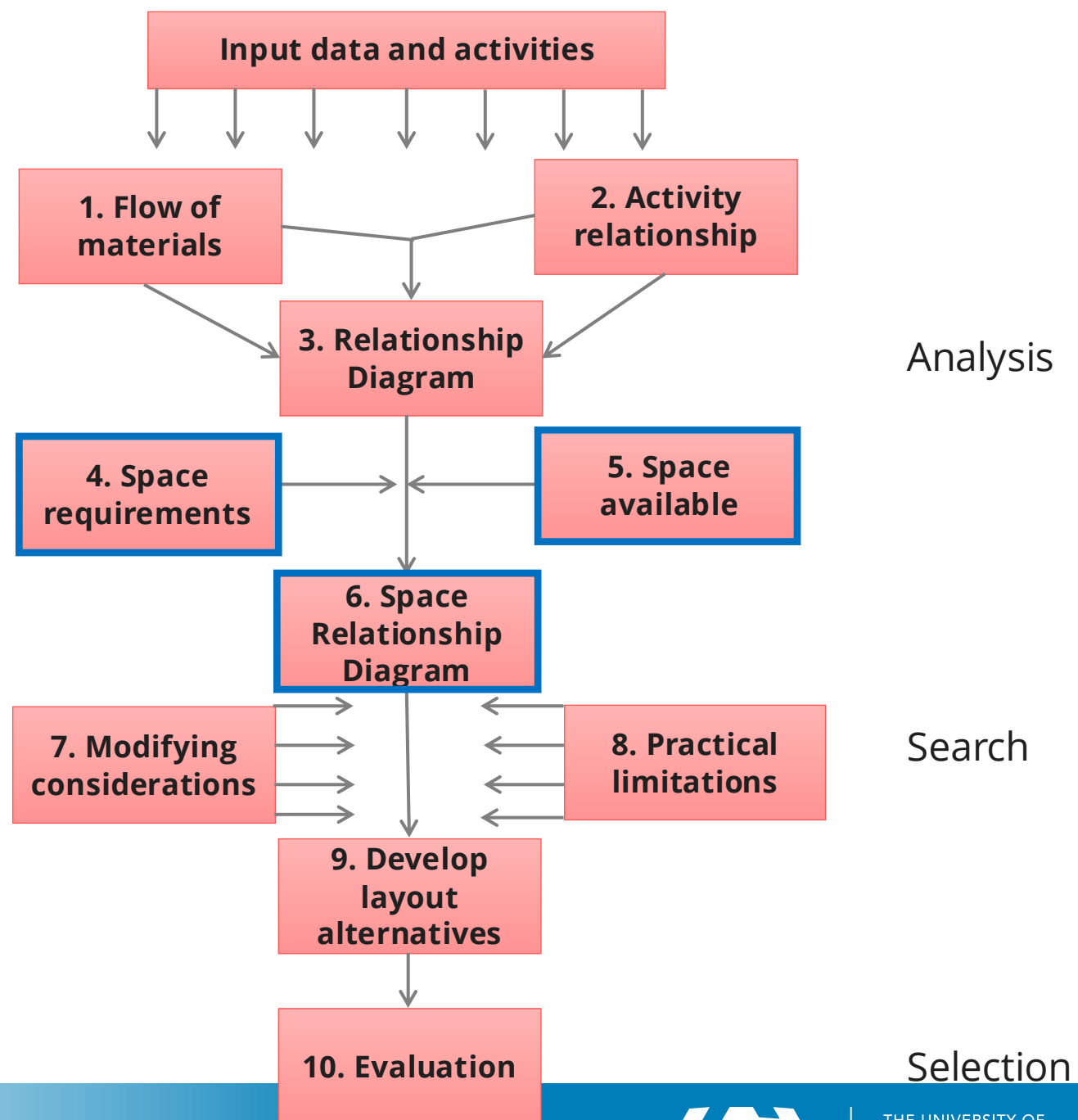
How many relationships?

- $n(n-1)/2$
- 10 departments → 45 relationships; 50 departments → 1225 relationships
- Large number of departments: iterative approach – group departments that are likely to be located together (“pseudo-departments”) and create activity relationship chart

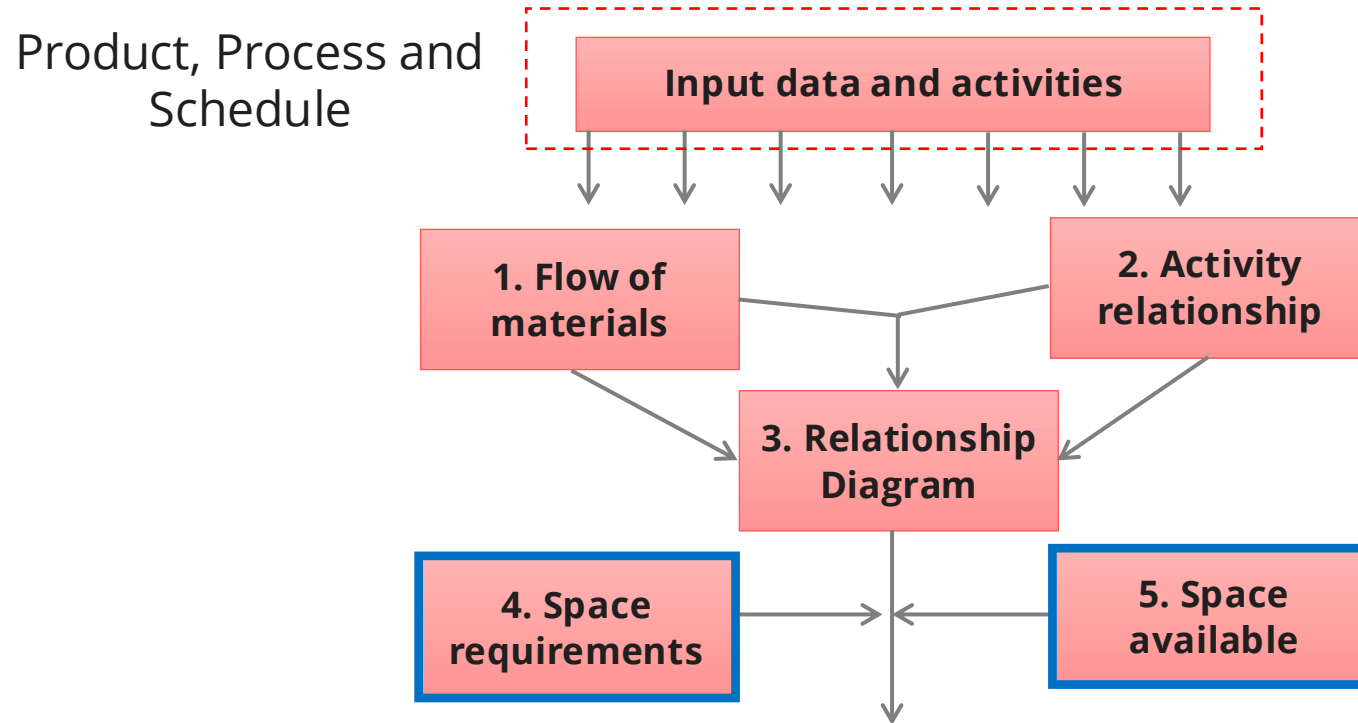
For facilities planner, activity relationships translate as proximity requirement

Can create activity relationship charts for material flow, personnel flow, information flow, organizational, control, environmental process

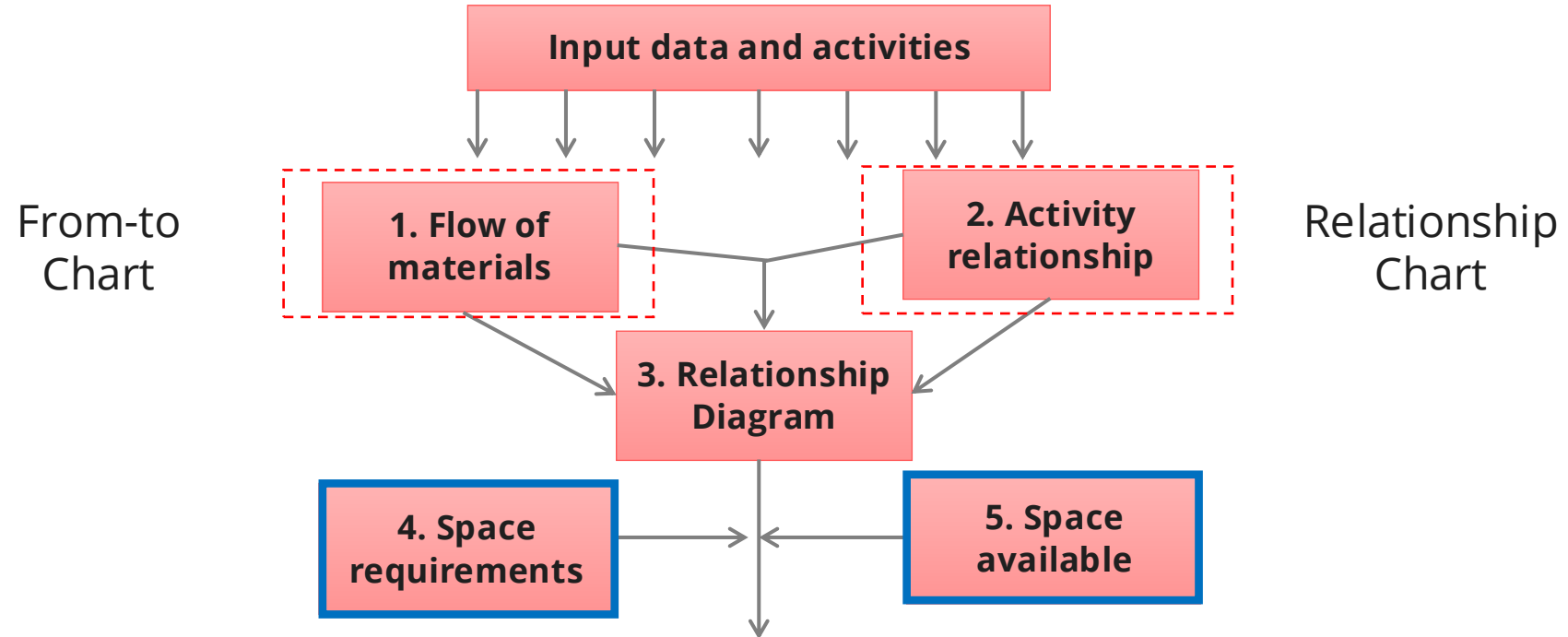
Systemic Layout Planning (SLP) Procedure



Systemic Layout Planning Procedure



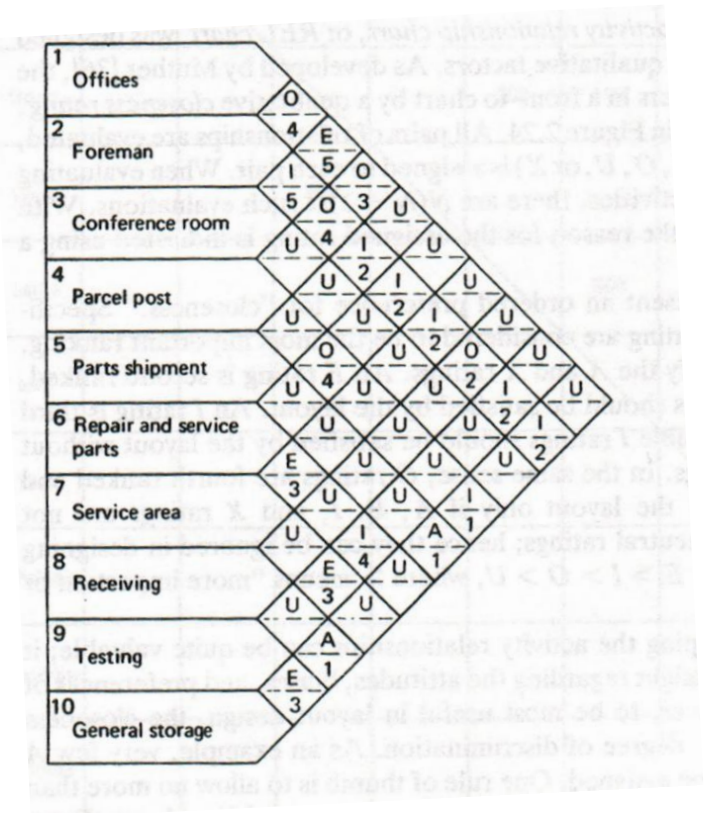
Systemic Layout Planning Procedure



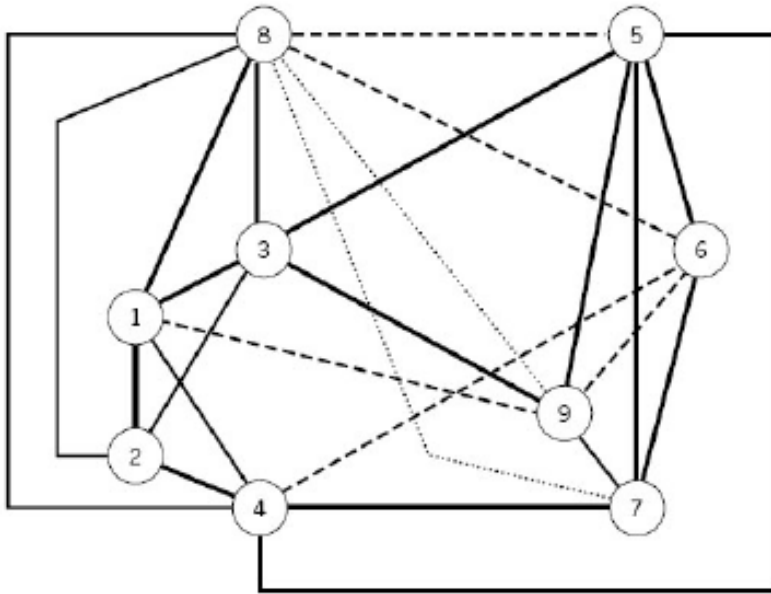
Space Planning

Given activity relationship diagram and space relationship diagram, space requirement is given in table.

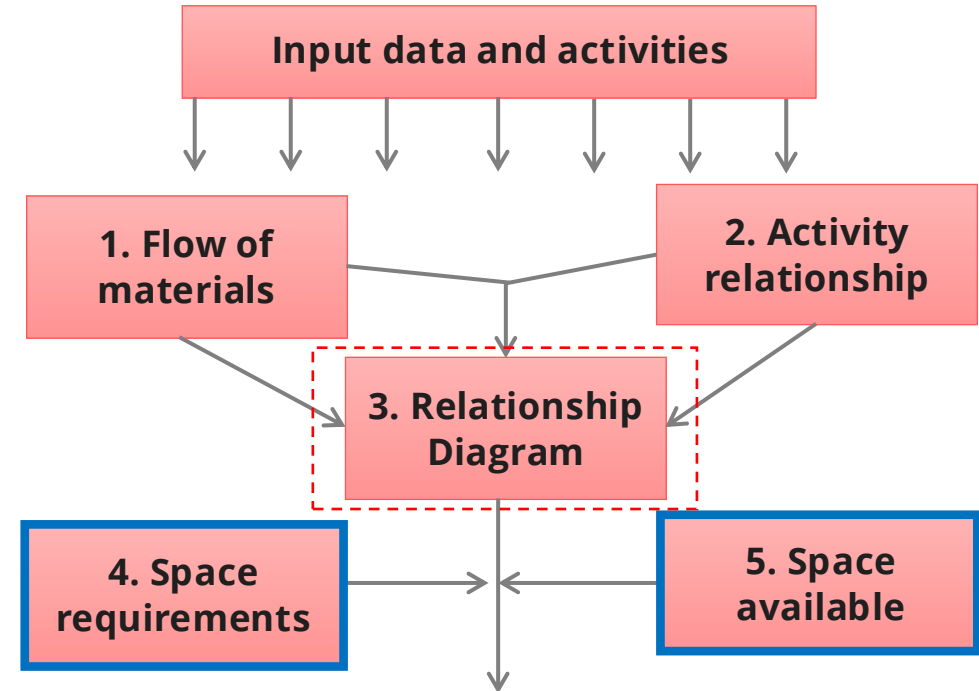
#	Department name	Area (ft ²)	# Unit Squares	Template dimensions
1	Office	1200	48	6X8
2	Foreman	150	6	2X3
3	Conference Room	300	12	3X4
4	Parcel Post	400	16	4X4
5	Parts Shipment	600	24	4X6
6	Repair, Service Parts	300	12	3X4
7	Service Area	900	36	6X6
8	Receiving	600	24	4X6
9	Testing	1000	40	5X8
10	General Storage	3500	120	10X12



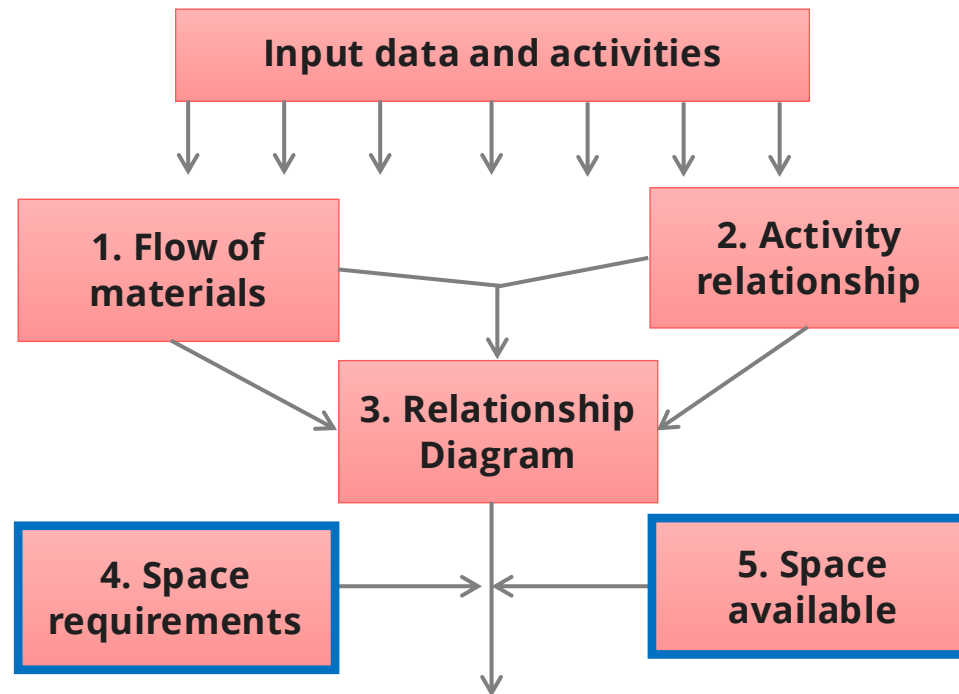
Systemic Layout Planning Procedure



Relationship diagram



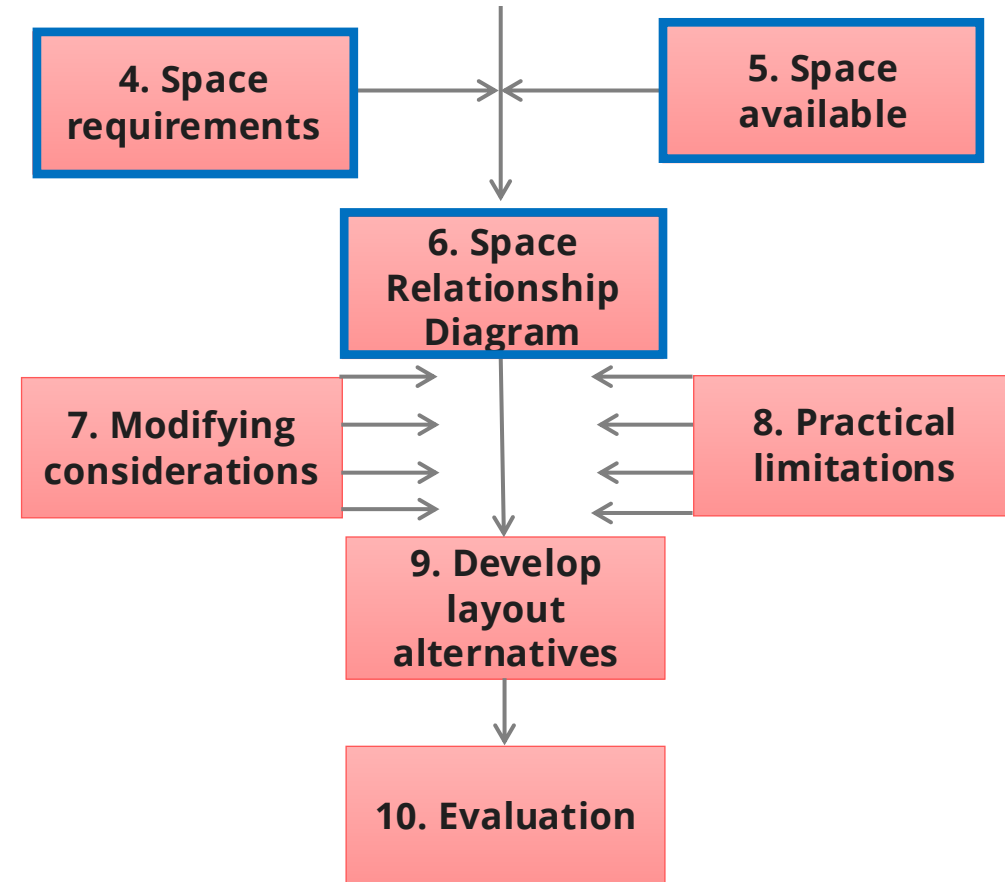
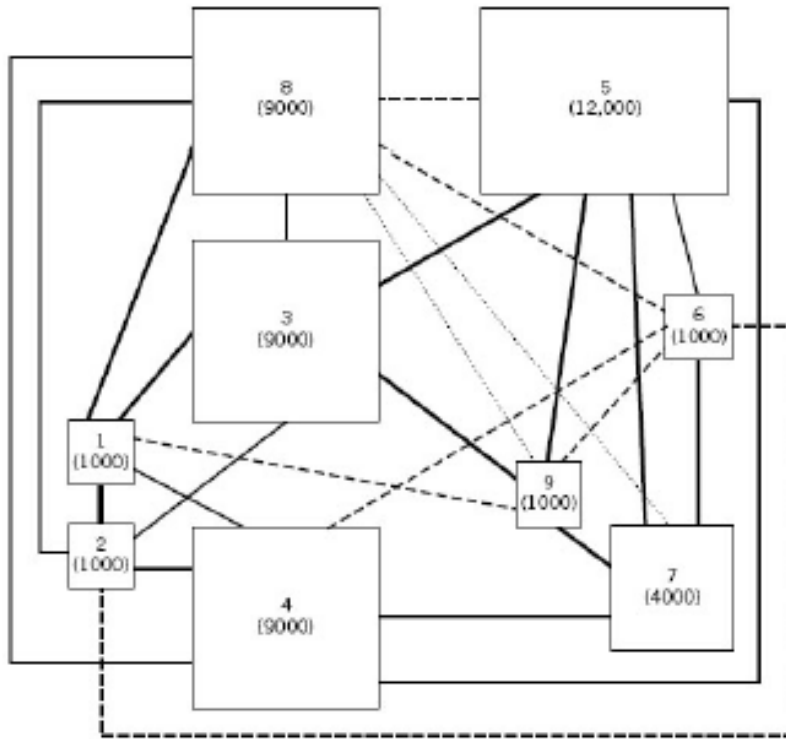
Systemic Layout Planning Procedure



Value	Closeness	Line code	Numerical weights
A	Absolutely necessary	=====	16
E	Especially important	=====	8
I	Important	=====	4
O	Ordinary closeness OK	=====	2
U	Unimportant	=====	0
X	Undesirable	=====	-80

Now let us take one step closer to develop out layout and focus on the space requirements.

Systemic Layout Planning Procedure



Space Planning

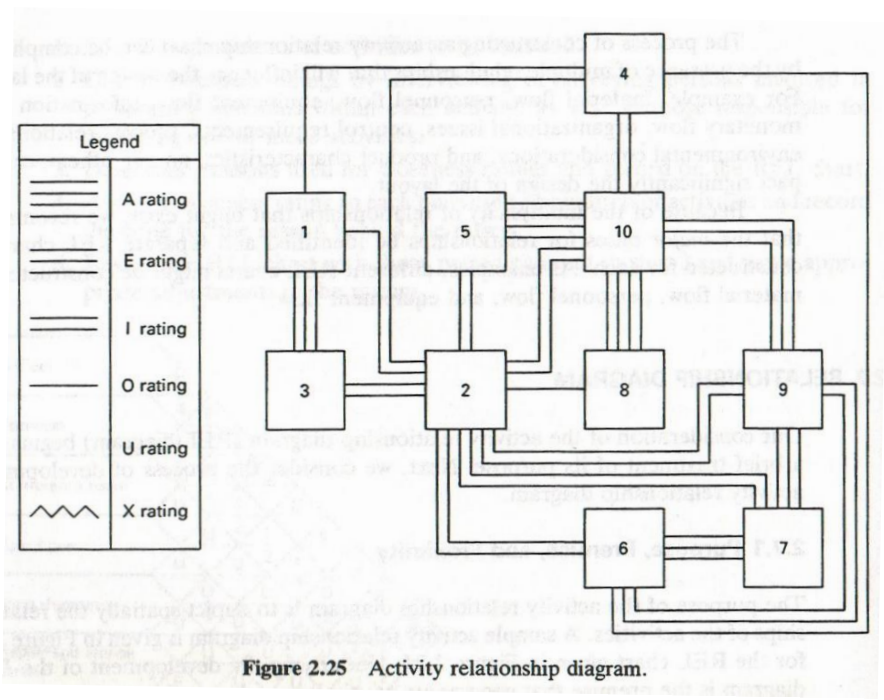


Figure 2.25 Activity relationship diagram.

#	Department name	Area (ft ²)	# Unit Squares	Template dimensions
1	Office	1200	48	6X8
2	Foreman	150	6	2X3
3	Conference Room	300	12	3X4
4	Parcel Post	400	16	4X4
5	Parts Shipment	600	24	4X6
6	Repair, Service Parts	300	12	3X4
7	Service Area	900	36	6X6
8	Receiving	600	24	4X6
9	Testing	1000	40	5X8
10	General Storage	3500	120	10X12

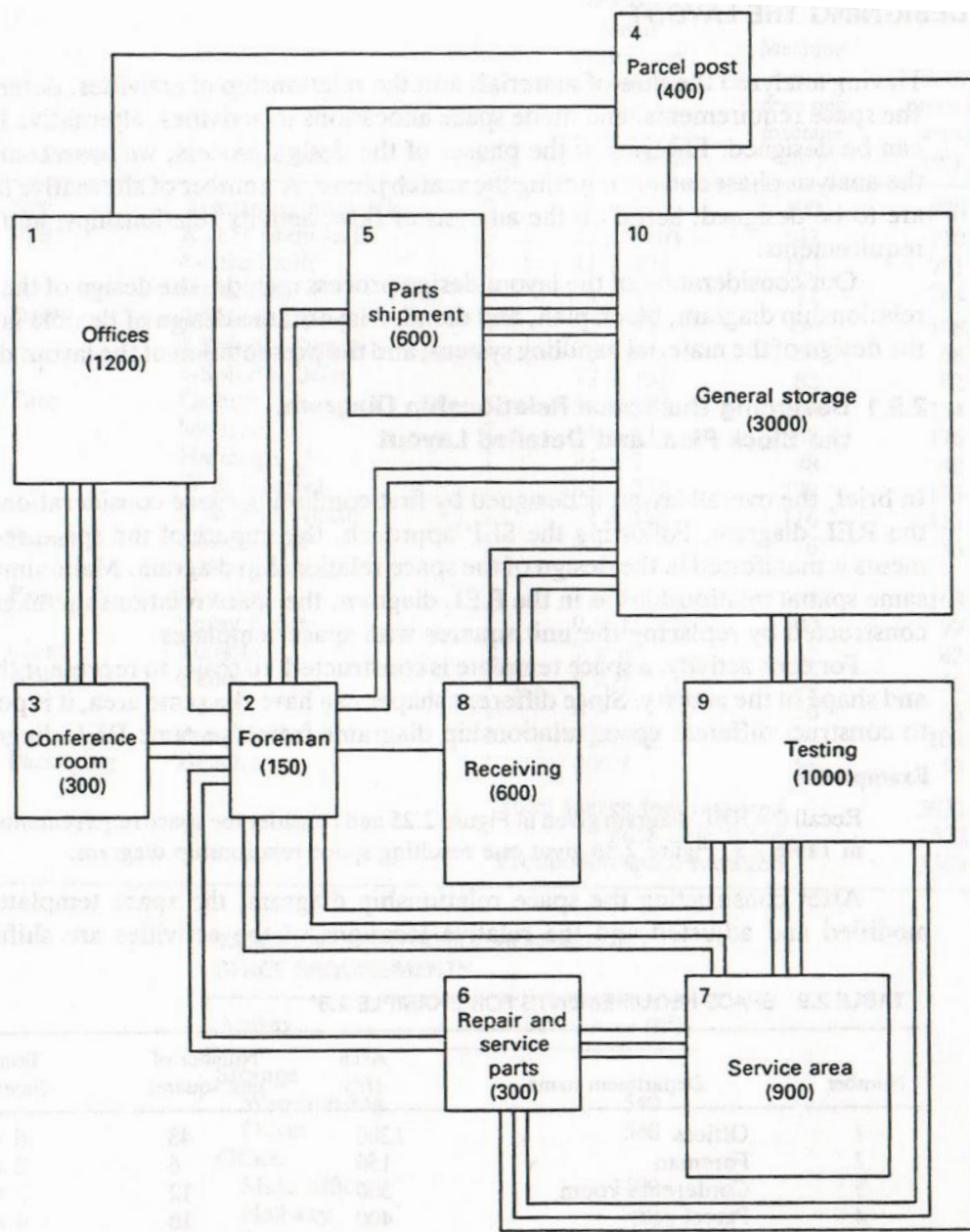
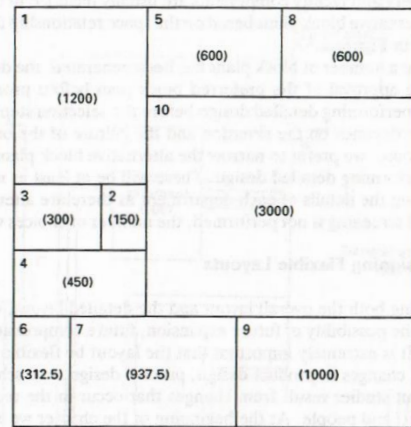


Figure 2.56 Space relationship diagram.

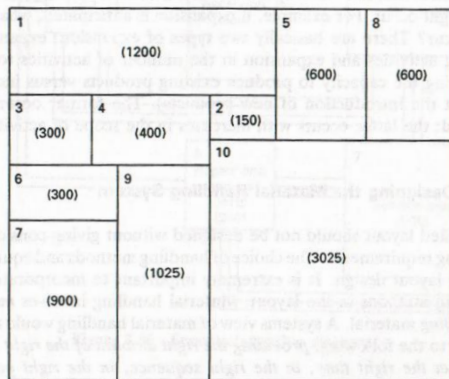
#	Department name	Area (ft ²)	# Unit Squares	Template dimensions
1	Office	1200	48	6X8
2	Foreman	150	6	2X3
3	Conference Room	300	12	3X4
4	Parcel Post	400	16	4X4
5	Parts Shipment	600	24	4X6
6	Repair, Service Parts	300	12	3X4
7	Service Area	900	36	6X6
8	Receiving	600	24	4X6
9	Testing	1000	40	5X8
10	General Storage	3500	120	10X12

Space Planning

- Given space relationship diagram with space requirement, alternatives can be generated

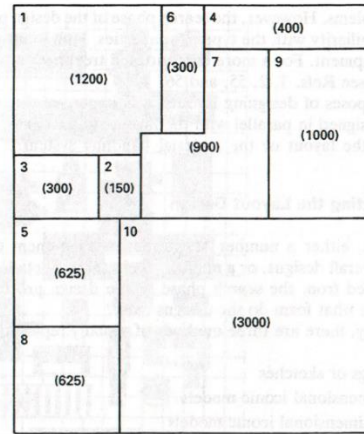


(a) Total area = 8,550 sq. ft.

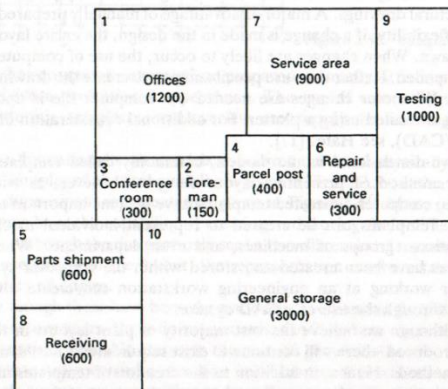


(b) Total area = 8,500 sq. ft.

Figure 2.57 Alternative block plans.



(c) Total area = 8,500 sq. ft.



(d) Total area = 8,450 sq. ft.

Figure 2.57 (Continued)

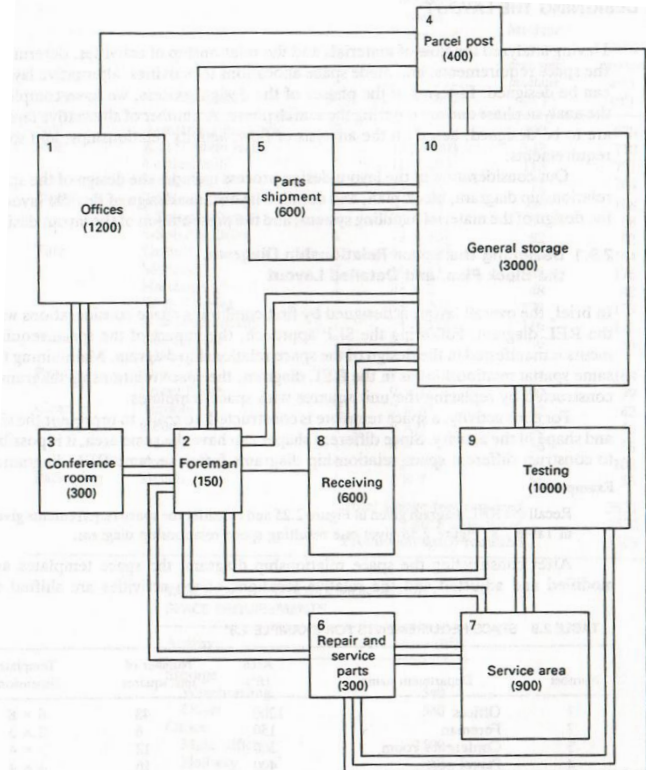


Figure 2.56 Space relationship diagram.

Space Planning

- Optimal design:
 - Optimize use of space within feasible range
 - Criteria of optimality:
 - Efficiency & productivity, material handling, cost, operator safety & fatigue.
- Strategic planning: for future expansions/changes (5-10 years)
- Parkinson's law: things will expand to fill all available capacity sooner than you plan
- Ground-up approach: first form workstations then departments

Space Requirements

- **Work Areas:** equipment, worker, material storage, material handling, tool rooms & cribs, maintenance, quality control & inspection, supervision, offices
- **Non-Work Areas:** shipping/receiving, and food, medical & convenience facilities



Space Requirements

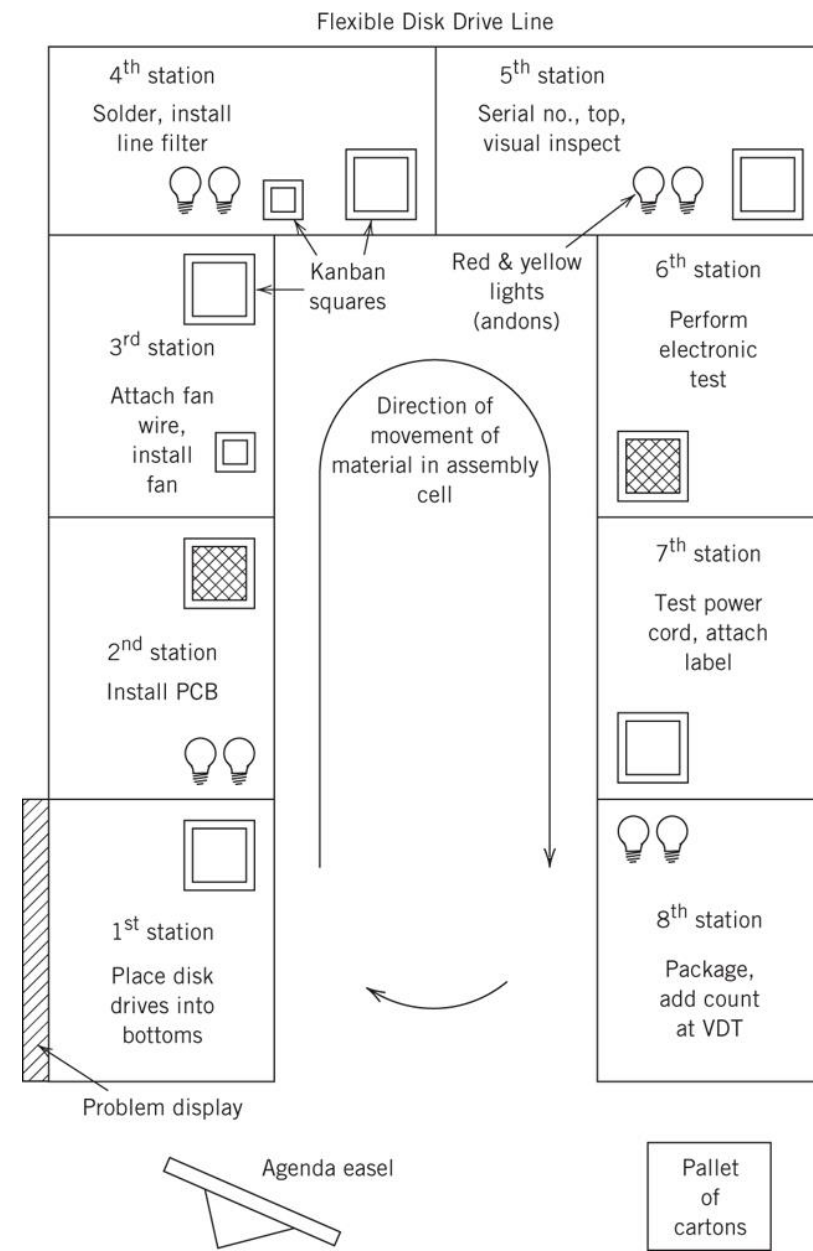
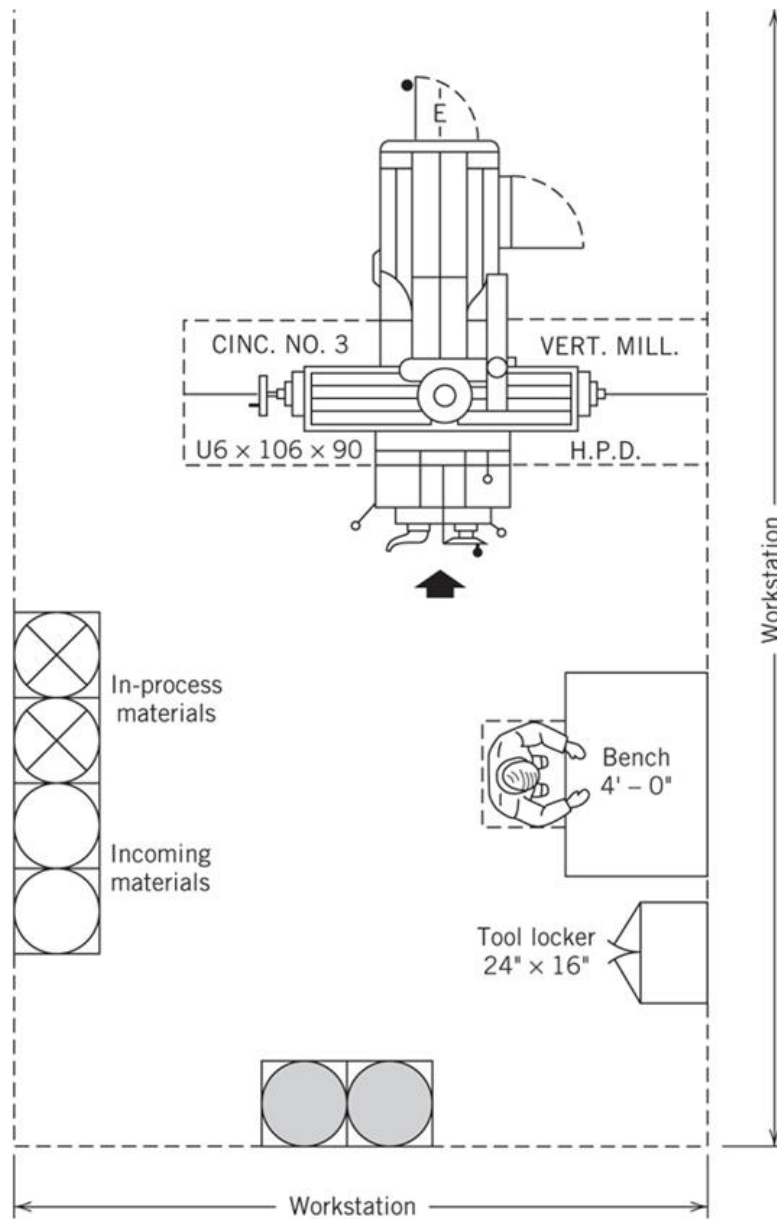
- Efforts to improve operational efficiency and costs will also impact current and future space needs
- Generally, space requirements are being reduced by:
 - Products are delivered to points of use in smaller lots and unit load sizes
 - Decentralized storage areas are located at points of use
 - Less inventories are carried
 - More efficient layout arrangements (i.e., manufacturing cells)
 - Companies downsizing
 - Shared offices and telecommuting
 - Outsourcing

Space Requirements for Workstations

- Equipment
 - Static width, depth and height
 - Travel: vertical, left, right, forward and backward
 - Maintenance and plant service area
- Material
 - Receiving and storing materials
 - Work-in-process
 - Shipping; input and output buffers
 - Storing and shipping waste and scrap
 - Tools, fixtures, jigs, dies, spare parts, maintenance materials
- Personnel
 - Operator
 - Material handling
 - Operator ingress and egress
- Plant service:
 - Utility lines, vent hood, air conditioning.....

General Guidelines for Workstation Design

- Workstations should be designed so that:
 - Operators can pick up and discharge materials without walking or making long or awkward reaches
 - There is efficient and effective utilization of the operator
 - Minimize time spent manually handling materials
 - Maximize operator safety, comfort, and productivity
 - Minimize hazards, fatigue, and eye strain



Departmental Space Requirements

- Service requirements
 - Not simply the sum of areas of individual workstations
 - Must consider: tooling/fixtures, maintenance and plant services, housekeeping items, storage, operators, spare parts, Kanban boards, and material handling
- Must approximate aisle requirements at this stage (finalize later once layout is determined)

Space Requirements for Departments

- Combine if possible spaces for plant services, operator ingress and egress, and material storage
- Provide aisle space for material handling

Largest load	Aisle allowance percentage
Less than 6 ft ²	5-10%
Between 6 and 12 ft ²	10-20%
Between 6 and 12 ft ²	20-30%
Greater than 18 ft ²	30-40%

Space Requirements for Departments

- Aisle arrangements
 - Recommended aisle width for unidirectional traffic (in table)
 - Avoid curves, non right angle intersections, aisle along outside walls
 - Use columns to frame/border aisles
 - Use straight paths and lead to doors

Type of Flow	Aisle Width (feet)
Tractor	12
3-ton forklift	11
2-ton forklift	10
1-ton forklift	9
Narrow aisle truck	6
Manual platform truck	5
Personnel	3
Personnel with door opening into aisle from one side	6
Personnel with door opening into aisle from two sides	8

Space Requirements for Departments

- Example 1: ABC company consists of total of 13 machines that perform turning operation: 5 turret lathes machines, 6 automatic screw, 2 chucker machines. Aisle space allowance of 13% is used.



Type	#	"foot-prints"	Unit material storage requirement
Turret Lathes	5	4X12	20 ft ²
Automatic Screw	6	4X14	40 ft ²
Chucker	2	5X6	50 ft ²
Personnel	13	4X5	

Each people work on one machine

Space Requirements for Departments

Type	#	"foot-prints"	Unit material storage requirement
Turret Lathes	5	4X12	20 ft ²
Automatic Screw	6	4X14	40 ft ²
Chucker	2	5X6	50 ft ²
Personnel	13	4X5	

Type	#	Area (ft2)			
		Equipment	Material	Personnel	Total
Turret Lathes	5				
Automatic Screw	6				
Chucker	2				

Net area required:1336
 13% Aisle allowance: 173.68
 Total: 1509.68

Personnel Requirements

- Personnel Requirements Planning includes:
 - Employee parking, locker rooms, restrooms, food services, drinking fountains, and health services
 - 1990 Americans with Disabilities Act (ADA)
 - Disabilities and Business Technical Assistance Centers (DBTAC) offer up-to-date advice to plan for disability requirements

Personnel Philosophies

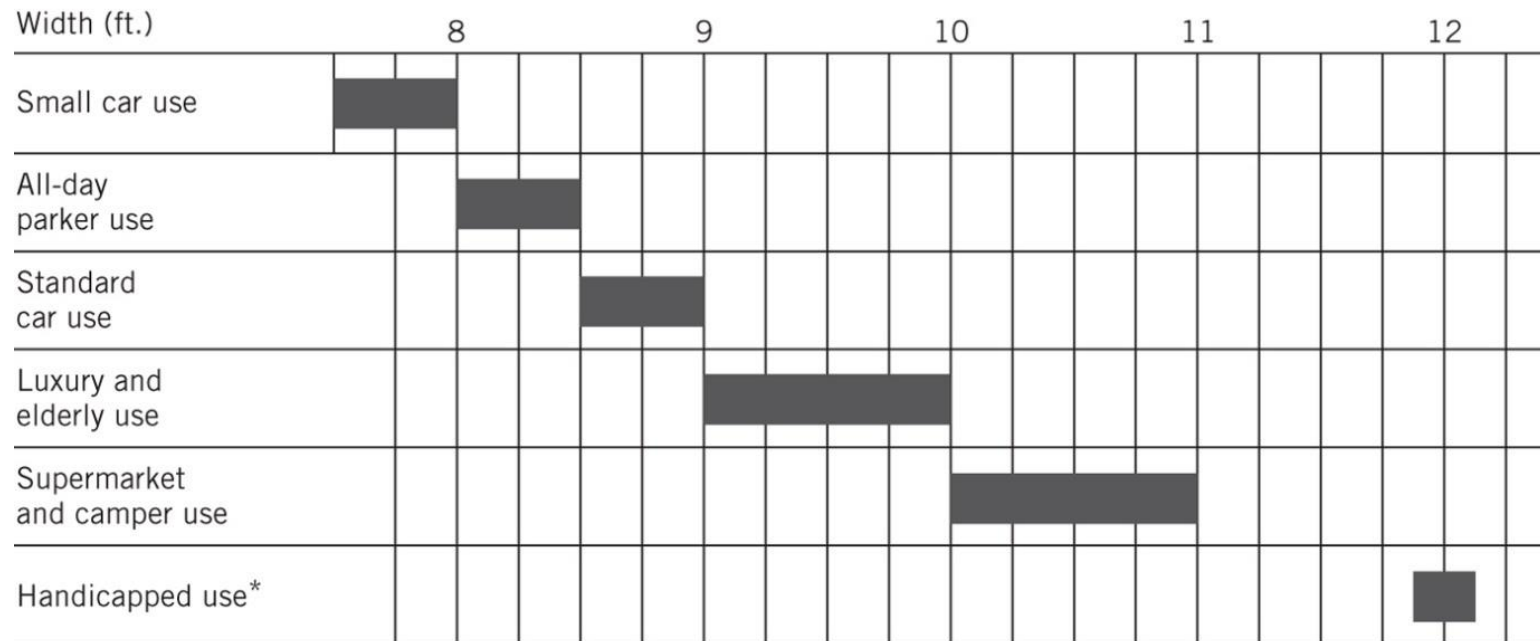
- “Our firm is responsible for our employees from the moment they leave their homes until they return. We must provide adequate methods of getting to and from work”
- “Employees should earn their parking locations; all spaces should be assigned to specific individuals”
- “Employees spend one-third of their lives within our facility; we must help them enjoy working here”
- “A happy worker is a productive worker”
- “A hot lunch makes works more productive since it supplies them energy”
- “Workers who do not feel well are unsafe workers; we should provide medical care to maintain health”

Personnel Philosophies

- “Our company has an obligation to our personnel; we will make all our facilities ADA compliant”
- “Except for individuals in private offices, no one is allowed to smoke in our building”
- “Our employees work hard; the least we can do is provide a place for them to unwind after hard day’s work. Employees who play together will work better together”
- “Personnel considerations are of little importance in our facility. We pay people to work, not to have a good time”

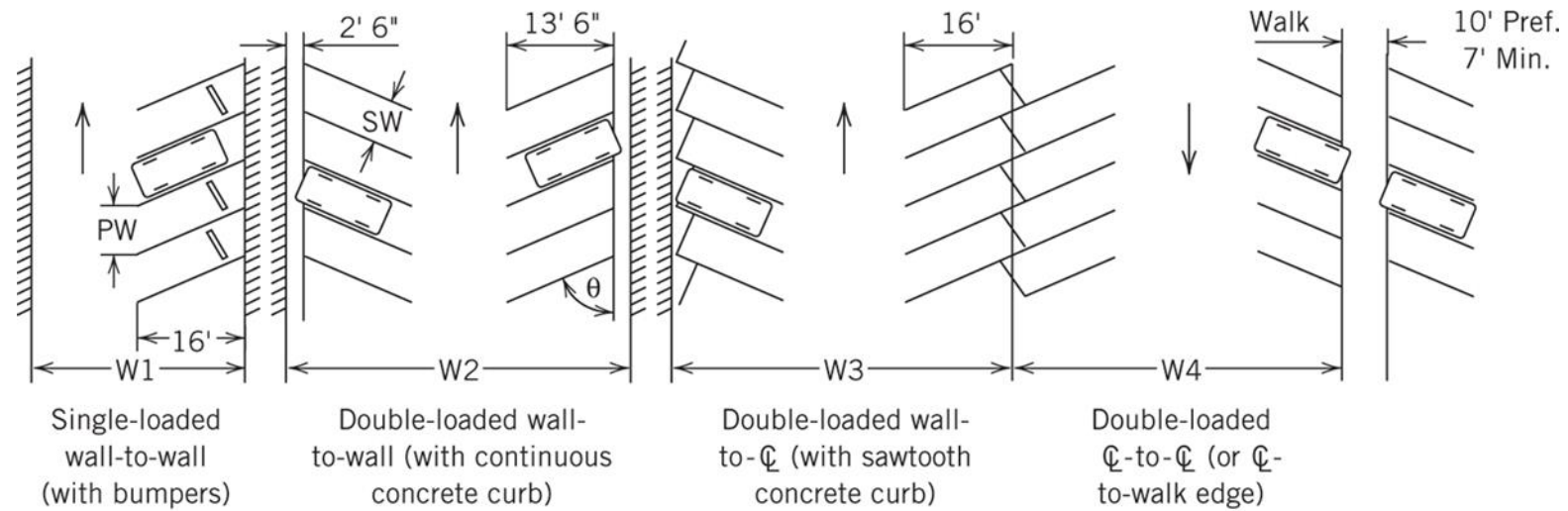
Employee Parking

- Stall widths range from 5.5'x12' to 9.5'x19' depending on car type
- Total area required depends on size of individual stalls, parking angle, and aisle width



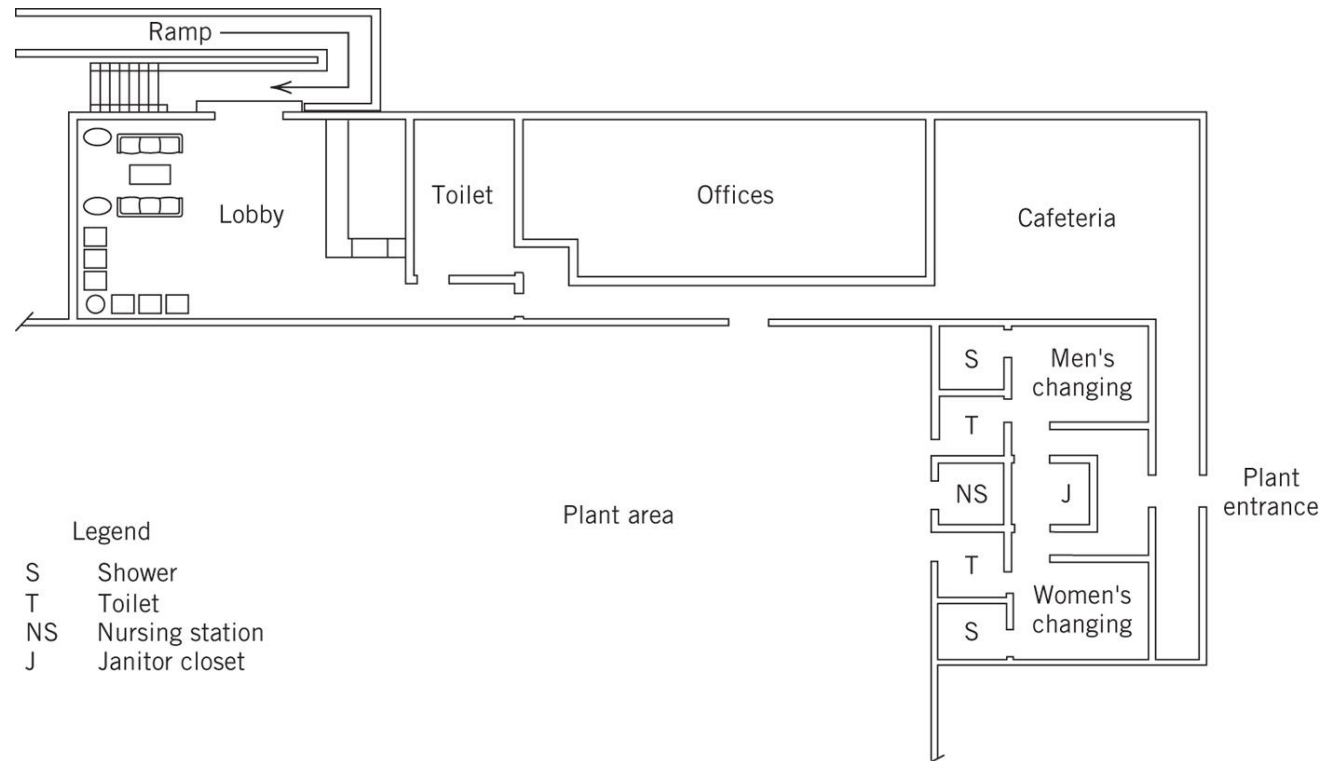
* Minimum requirements = 1 or 2 per 100 stalls or as specified by local, state, or federal law; convenient to destination.

Parking Configurations



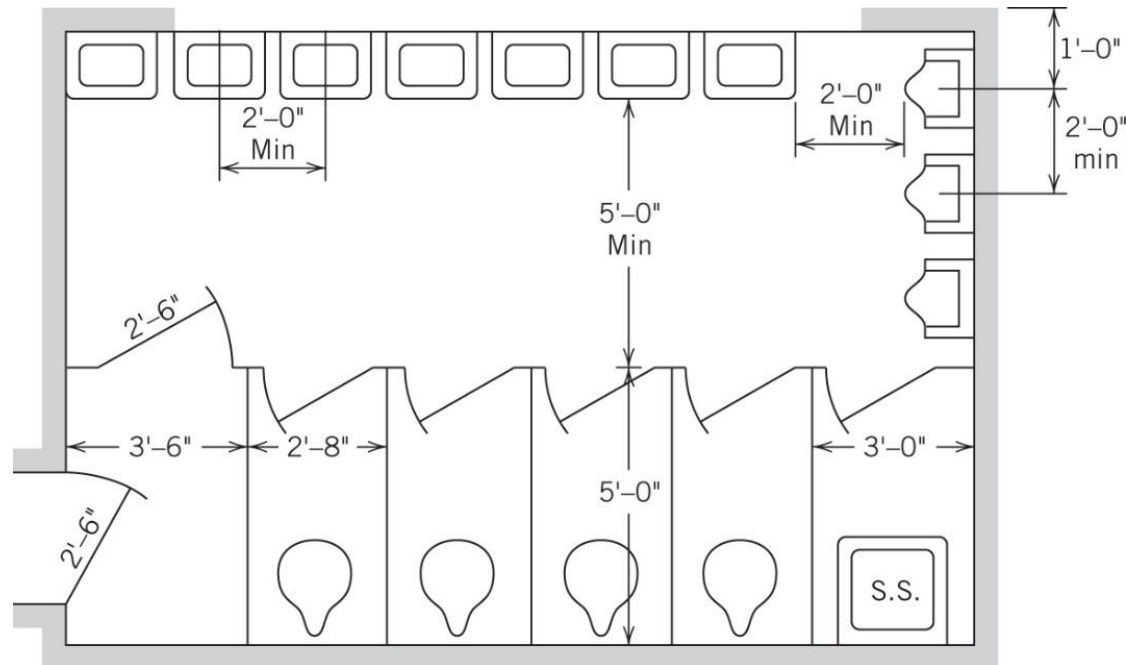
Storage of Personal Belongings

- Storage of Personal Belongings
 - Locker rooms: 6 ft² per person
 - Between entrance and work area along outside wall



Restrooms

- Rest Rooms
 - Within 200 ft of every permanent workstation
 - Decentralized restrooms provide greater convenience.
 - Must comply with local zoning regulations
 - 15 ft² for toilet, 6 ft² for urinal, 6 ft² for sink, 15 ft² for entrance



Business, Mercantile, Industrial Other Than Foundry and Storage					
Water Closets		Employees		Lavatories	
1		1–15		1	
2		16–35		2	
3		36–55		3	
4		56–80		4	
5		81–110		5	
6		111–150		6	
7		151–190		7	
				8	
One additional water closet for each 40 employees in excess of 190.			One additional lavatory for each 30 employees in excess of 175.		
Industrial, Foundries, and Storage					
Water Closets		Employees		Lavatories	
1		1–10		1	
2		11–25		2	
3		26–50		3	
4		51–80		4	
5		81–125		5	
One additional water closet for each 45 employees in excess of 125.			One additional lavatory for each 25 employees in excess of 65.		
Assembly, Other Than Religious, and Schools					
Water Closets	Occupants	Urinal	Male Occupants	Lavatories	Occupants
1	1–100	1	1–100	1	1–100
2	101–200	2	101–200	2	101–200
3	201–400	3	201–400	3	201–400
4	401–700	4	401–700	4	401–700
5	701–1100	5	701–1100	5	701–1100
One additional water closet for each 600 occupants in excess of 1100.		One additional urinal for each 300 occupants in excess of 1100.		One additional lavatory for each 1500 occupants in excess of 1100. Such lavatories need not be supplied with hot water.	

Food Services

- Location within facility (guidelines)
 - Within 100 feet of all permanent employee workstations
 - If employees must travel more than 1000 feet consider decentralized food services
 - Centrally located and positioned so that distance from farthest workstation is minimized
 - Located to allow delivery of food and trash pickup
 - Allow employees an outside view while eating
 - Adequate ventilation so odors and exhaust do not interfere with other activities

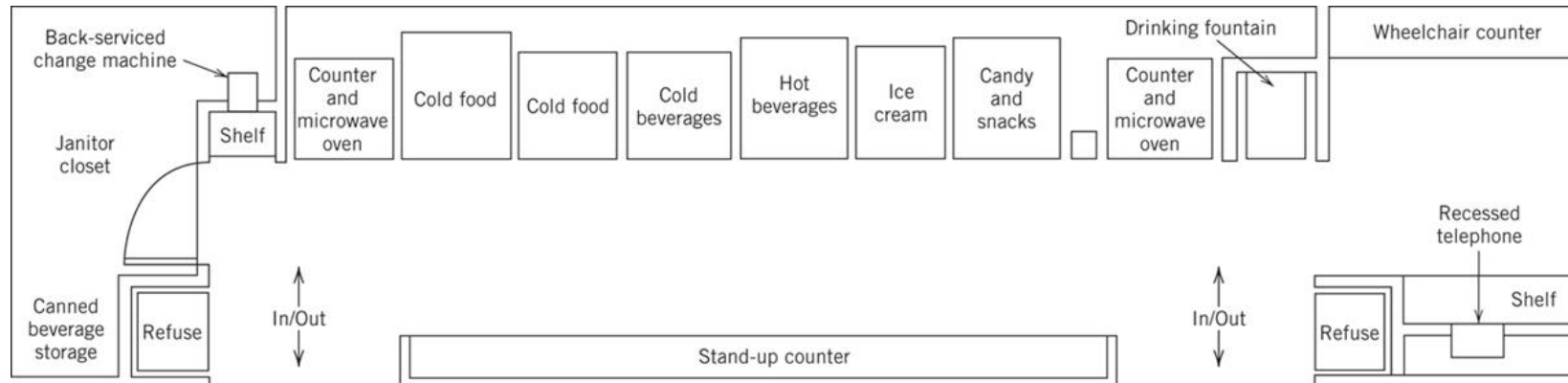
Food Services

- Occupational Safety & Health Act (1970) forbids consumption of food where toxic substances exist
- Typically, 1/3 time for preparing and obtaining meal, 2/3 time for eating
 - For 30-min meal break, 20 min dining shift
 - For 45-min meal break, 30 min dining shift

Beginning of Lunch Break	Time Sat Down in Chair	End of Lunch Break
11:30 A.M.	11:40 A.M.	12:00 Noon
11:50 A.M.	12:00 Noon	12:20 P.M.
12:10 P.M.	12:20 P.M.	12:40 P.M.
12:30 P.M.	12:40 P.M.	1:00 P.M.

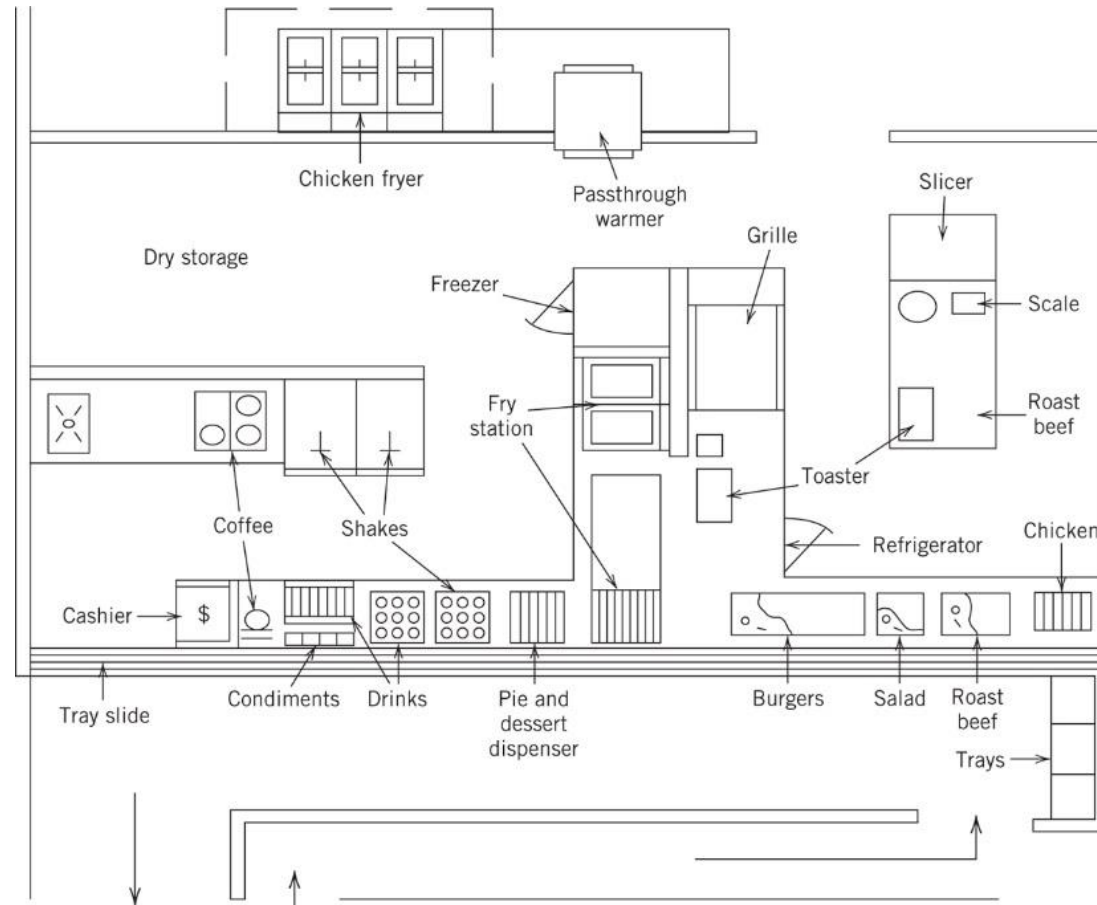
Food Service

- Examples:
 - Dining away from facility, vending machines, serving line & cafeteria, full kitchen & cafeteria
 - Each type has corresponding space requirement given # employees and classification of facilities (e.g., commercial, industrial, banquet)



Vending Area Layout

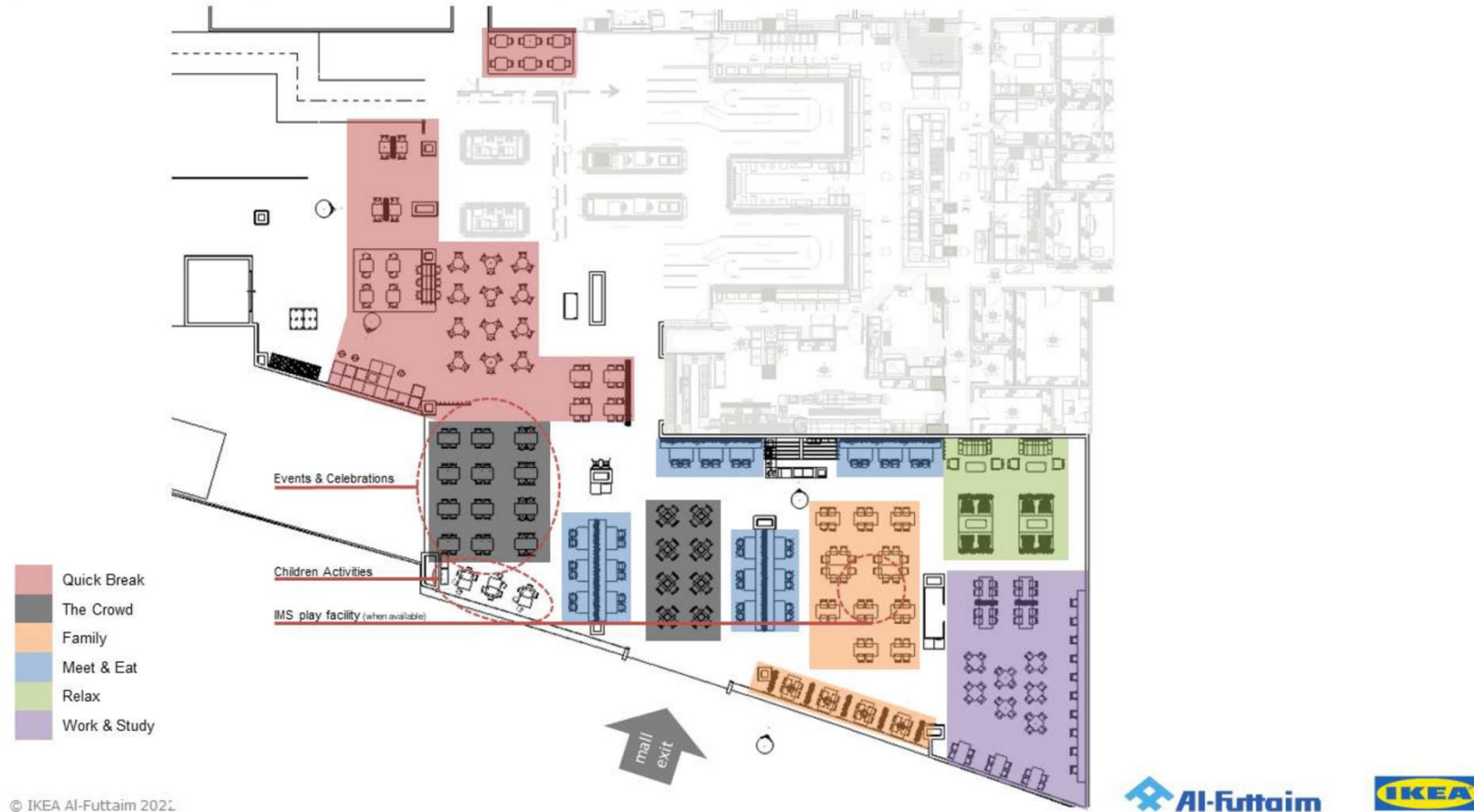
Food Service



Caterer Area & Serving Line Layout

Food Service

| RESTAURANT & CAFE | IKEA MOA STORE | Jan 2024 | LAYOUT ZONING |



Source

Food Services

- Facility Planning Considerations
 - General: Firm's view of and policies regarding food service, number of employees
 - Kitchen facilities: Total number of meals to be served
 - Cafeteria facilities: Maximum number of employees to eat in area at one time, style of tables (rectangular, square, circular)

Number of Meals Served	Area Requirements (ft ²)
100–200	500–1000
200–400	800–1600
400–800	1400–2800
800–1300	2400–3900
1300–2000	3250–5000
2000–3000	4000–6000
3000–5000	5500–9250

Full Kitchen Space Requirements

Classification	Square Footage Allowance per Person
Commercial	16–18
Industrial	12–15
Banquet	10–11

Cafeteria Space Requirements

Food Services

- Exercise 3: An industrial facility with 600 employees where they eat in three 30-min shifts. How much space should be planned for a cafeteria with serving lines and a full kitchen?

Cafeteria: 600 people 3 shifts, $12 * 200 = 2400 \text{ ft}^2$

Serving line: "Typical industrial serving line requires 300 ft^2 and can serve 7 employees per minute"

In 10 minutes (1/3 of lunch shift) can handle 70 people → 3 lines needed: 900 ft^2

2100 ft^2 for full kitchen space for serving 600

therefore **5400 ft^2**

Classification	Square footage per person
Commercial	16-18
Industrial	12 15
Banquet	10-11

# meals	Area Requirement (ft^2)
100-200	500-1000
200-400	800-1600
400-800	1400-2800 2100
800-1300	2400-3900



Examples:

XYZ Inc. has a facility with six departments (A, B, C, D, E, and F). A summary of the processing sequence for 10 products and the weekly production forecasts for the products are given in the tables below.

- Develop the from-to chart based on the expected weekly production.
- Develop a block layout using SLP.

Product	Processing Sequence	Weekly Production
1	ABCDEF	960
2	ABCBEDCF	1200
3	ABCDEF	720
4	ABCEBCF	2400
5	ACEF	1800
6	ABCDEF	480
7	ABDECBCF	2400
8	ABDECBCF	3000
9	ABCDF	960
10	ABDEF	1200

Dept.	Dimension
A	40' * 40'
B	45' * 45'
C	30' * 30'
D	50' * 50'
E	60' * 60'
F	50' * 50'

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4	ABCEBCF	2400
5	ACEF	1800
6	ABCDEF	480
7	ABDECBF	2400
8	ABDECBF	3000
9	ABCDF	960
10	ABDEF	1200

Dept.	A	B	C	D	E	F
A	-					
B		-				
C			-			
D				-		
E					-	
F						-

A company produces four products in its production facility. The facility has four departments labeled A–D. The routing information, production quantities, and volumes for each product are given in the table below.

Product	Production Quantities (per day)	Routing	Volume (compared to Product 1)
1	15	B-A-D-C-A	1
2	20	A-D-C-B	1
3	14	A-C-D-B	2
4	10	B-A-C-D	0.5

A company produces four products in its production facility. The facility has four departments labeled A–D. The routing information, production quantities, and volumes for each product are given in the table below.

Product	Production Quantities (per day)	Routing	Volume (compared to Product 1)
1	15	B-A-D-C-A	1
2	20	A-D-C-B	1
3	14	A-C-D-B	2
4	10	B-A-C-D	0.5

	A	B	C	D
A	-			
B		-		
C			-	
D				-